

Submerged Aquatic Vegetation Indicator Quantile Report

SEACAR Analysis

Last compiled on 22 October, 2025

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Overview

Purpose

The purpose of the indicator quantiles is to flag records that are “unusual” relative to all of the data in the DDI for a given indicator in order to facilitate QA/QC. They are not used to filter any of the data for SEACAR analyses, and the presence of a LowerQuantile or UpperQuantile flag on a DDI record alone does not necessarily indicate there is any issue with the record (neither does the absence of a LowerQuantile or UpperQuantile flag necessarily mean that a data record is correct).

Relevant file locations

Current values can be found in the “LowQuantile” and “HighQuantile” columns of the “Ref_Parameters” worksheet.

The R script described below and the output file can be found in the *FloridaSEACAR IndicatorQuantiles* repository on GitHub:

- <https://github.com/FloridaSEACAR/IndicatorQuantiles>

Process steps

IQ_Report_Render.R & IQ_Report.Rmd

1. The *IQ_Report_Render.R* script lists all files in a given directory and filters it to a list of DDI exports to evaluate considering a list of parameters to skip (user-defined).
2. User sets the desired upper and lower quantile thresholds, as well as a number of standard deviations away from the mean to use for the calculations.
3. User sets the string value(s) in the DDI exports that should be considered as NA values.
4. The remainder of the script loops through the file list, returning the values listed below and binding them together by row into a single Excel spreadsheet that is saved to the User’s working directory.
5. For each habitat included in the User’s working directory a PDF report will be created in the “output” folder using *IQ_Report.Rmd*, which provides an overview of questionable / flagged values.
6. In addition to the PDF reports, each habitat will provide a .txt data output file in the “output/data” folder containing questionable values.

Summary

The following quantile thresholds are used for flagging “questionable” values:

- Lower quantile: **0.001**
- Upper quantile: **0.999**

Included Indicators and Parameters and the files used in this analysis:

All_SAV_Parameters-2025-Sep-04.txt

Indicator: Percent Cover

- Braun Blanquet Score
- Modified Braun Blanquet Score
- Percent Cover
- Percent Occurrence
- Presence/Absence
- Shoot Count

Summary Tables

q_low: Value corresponding to the qval_low quantile for the parameter in the DDI export.

q_high: Value corresponding to the qval_high quantile for the parameter in the DDI export.

mean: Mean value for the parameter in the DDI export.

n_tot: Total number of records in the DDI export for the parameter.

n_q_low: Number of records in the DDI export that are below q_low for the parameter.

n_q_high: Number of records in the DDI export that are above q_high for the parameter.

pct_flagged: Proportion of total records in the DDI export for the parameter which have been flagged as above q_high, or below q_low.

Indicator: **Percent Cover**

Table 1: Indicator Quantile Overview

ParameterName	q low	q high	mean	n tot	n q low	n q high	pct flagged
Braun Blanquet Score	0	5	0.30	5019307	4	5	0
Modified Braun Blanquet Score	0	5	1.96	52493	0	1	0
Percent Cover	0	100	4.05	1035818	0	46	0
Percent Cover	0	100	4.26	900385	0	45	0
Percent Occurrence	0	100	9.36	1193687	0	3	0
Percent Occurrence	0	100	9.76	966429	0	3	0
Presence/Absence	0	1	0.19	6374583	0	0	0

Low Quantile

Indicator: Percent Cover

Braun Blanquet Score

Table 2: Flagged Values - Low Indicator Quantile: **0**

RowID	ProgramID	ProgramLocationID	SampleDate	CommonIdentifier	ResultValue
94	296	276	2020-12-13	Thalassia testudinum	-0.69
89	296	505	2020-08-14	Thalassia testudinum	-0.30
21600	296	DRTO12	2020-07-30	Syringodium filiforme	-1.00
55957	296	DRTO12	2020-07-30	Calcareous green algae	-1.56

Programs containing flagged data:

296 - Florida Keys National Marine Sanctuary Seagrass Monitoring Project

There are no *Low* Quantile Flagged Values for Modified Braun Blanquet Score

There are no *Low* Quantile Flagged Values for Percent Cover

There are no *Low* Quantile Flagged Values for Percent Occurrence

There are no *Low* Quantile Flagged Values for Presence/Absence

There are no *Low* Quantile Flagged Values for Shoot Count

High Quantile

Indicator: Percent Cover

Braun Blanquet Score

Table 3: Flagged Values - High Indicator Quantile: **5**

RowID	ProgramID	ProgramLocationID	SampleDate	CommonIdentifier	ResultValue
10102693	4018	13M	2018-07-19	Anadyomene stellata	10
10128214	4018	13M	2018-07-19	Anadyomene stellata	7
10175177	4018	13M	2018-07-19	Anadyomene stellata	15
10156121	4018	68K	2018-06-28	Anadyomene stellata	6
10112663	4018	9P	2018-07-23	Anadyomene stellata	6

Programs containing flagged data:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program

Modified Braun Blanquet Score

Table 4: Flagged Values - High Indicator Quantile: **5**

RowID	ProgramID	ProgramLocationID	SampleDate	CommonIdentifier	ResultValue
1769092	559	KB06-2016	2016-06-30	Laurencia spp.	5.3

Programs containing flagged data:

559 - Northern Big Bend Seagrass Monitoring

Percent Cover

Table 5: Flagged Values - High Indicator Quantile: **100**

RowID	ProgramID	ProgramLocationID	SampleDate	CommonIdentifier	ResultValue
1612252	564	W1-06-12	2012-09-26	Total SAV	105
1720521	564	W1-12-11	2011-10-05	Total SAV	105
1750069	564	W1-16-17	2017-10-06	Total SAV	101
1623769	564	W1-17-11	2011-09-30	Total SAV	105
1739973	564	W1-18-14	2014-10-10	Total SAV	101
1685078	564	W1-22-19	2019-11-10	Total SAV	150
1705117	564	W1-22-19	2019-11-10	Total SAV	104
1756818	564	W1-23-11	2011-10-03	Total SAV	101
1805043	564	W1-24-11	2011-10-05	Total SAV	105
1740235	564	W1-25-15	2015-10-02	Total SAV	105
1485868	564	W1-26-12	2012-09-26	Total SAV	101
1623578	564	W1-27-11	2011-10-04	Total SAV	101
1720562	564	W1-27-11	2011-10-04	Total SAV	140
1666589	564	W1-28-12	2012-09-26	Total SAV	105
1667043	564	W1-28-12	2012-09-26	Total SAV	105
1705267	564	W1-28-19	2019-12-31	Total SAV	101
1577584	564	W1-31-15	2015-10-07	Total SAV	110
1663988	564	W1-31-15	2015-10-07	Total SAV	110
1672098	564	W1-31-15	2015-10-07	Total SAV	110
1685858	564	W1-31-19	2019-12-31	Total SAV	180
1686138	564	W1-34-15	2015-10-02	Total SAV	125
1521380	564	W1-35-19	2019-12-31	Total SAV	110
1521468	564	W1-35-19	2019-12-31	Total SAV	110
1521469	564	W1-35-19	2019-12-31	Total SAV	105
1679110	564	W1-35-19	2019-12-31	Total SAV	105
1685992	564	W1-35-19	2019-12-31	Total SAV	105
1705396	564	W1-35-19	2019-12-31	Total SAV	105
1548273	564	W1-40-19	2019-12-31	Total SAV	120
1509928	564	W2-01-12	2012-09-14	Total SAV	105
1504140	564	W2-04-11	2011-09-29	Total SAV	120
1645349	564	W2-04-12	2012-09-13	Total SAV	140
1642827	564	W2-04-15	2015-10-21	Total SAV	120
1592230	564	W2-2-19	2019-03-10	Total SAV	130
1676598	564	W2-2-19	2019-03-10	Total SAV	110
1517261	564	W2-5-19	2019-12-31	Total SAV	120
1590007	564	W2-5-19	2019-12-31	Total SAV	180
1683119	564	W2-9-19	2019-03-10	Total SAV	122
1697321	564	W2-ALT-01-17	2017-09-29	Total SAV	105
1506025	564	W3-03-15	2015-10-05	Total SAV	125
1506026	564	W3-03-15	2015-10-05	Total SAV	110
1577037	564	W3-03-15	2015-10-05	Total SAV	106
1637849	564	W3-03-15	2015-10-05	Total SAV	135
1663307	564	W3-03-15	2015-10-05	Total SAV	115
15351702	5027	CWA3	2024-05-28	Total seagrass	105
15120446	5027	DM08	2024-08-23	Total seagrass	105

Programs containing flagged data:

564 - Western Pinellas County Seagrass Monitoring
5027 - North Biscayne Bay Seagrass Loss Monitoring Program

Percent Occurrence

Table 6: Flagged Values - High Indicator Quantile: **100**

RowID	ProgramID	ProgramLocationID	SampleDate	CommonIdentifier	ResultValue
6891061	3013	IRLSG045	2021-08-12 09:15:00	Total seagrass	161
8129248	3013	IRLSG045	2021-08-12 09:15:00	Total seagrass	139
10226772	3015	LC7	2011-11-29 00:00:00	Thalassia testudinum	460

Programs containing flagged data:

3013 - Seagrass (SJRWMD)

3015 - SCCF Seagrass Surveys

There are no *High* Quantile Flagged Values for Presence/Absence

There are no *High* Quantile Flagged Values for Shoot Count